Project Phase 6: Full Automation - Infrastructure as Code & CI/CD

Due Date: End of Week 9 **Points:** 100

Scenario

The Riverside Cafe's cloud architecture is now robust, but the deployment process is not. All infrastructure has been created manually, which is slow, error-prone, and makes it difficult to create consistent environments for testing or expansion into new regions.

Your task is to solve these operational problems by moving from manual "click-ops" to a fully automated, Infrastructure as Code (IaC) approach.

Project Goals

1. **Codify Your Infrastructure.** Define your entire networking and application architectures from the previous phases using an IaC tool (e.g., AWS CloudFormation, Terraform).
2. **Version Control Your Code.** Store your IaC templates in a source code repository (e.g., AWS CodeCommit, GitHub).
3. **Build an Automated Pipeline.** Create a CI/CD pipeline that automatically deploys or updates your infrastructure whenever a change is pushed to the repository.
4. **Demonstrate Reproducibility.** Prove the power of your IaC setup by rapidly deploying a complete copy of your environment into a new region.

Technical Requirements

- You must create at least two separate IaC templates: a **network template** and an **application template**.
- The application template must reference outputs from the network template.
- Your pipeline must be triggered by a commit to your source code repository.

Submission Instructions

You will submit a single PDF document to Moodle. This document must contain all parts of the deliverable outlined below.

Deliverable Breakdown

- **Infrastructure as Code:** State which IaC tool and version control system you chose and briefly explain why. Include the full code for your network infrastructure template. Include the full code for your application infrastructure template. Explain how your application template references your network template and why separating them is a good architectural practice.
- **The CI/CD Pipeline:** Provide a diagram illustrating your CI/CD pipeline, showing how a `git push` triggers the source, build, and deploy stages. Provide screenshots of your pipeline's

configuration, showing the source and deployment stages. Show proof of a code change triggering an automatic deployment. Include screenshots of a `git push` command and your pipeline showing the corresponding commit message in a "Succeeded" state.

- **Disaster Recovery & Replication:** Describe the process of using your IaC templates to deploy a complete copy of your infrastructure into a *new region*. Provide a screenshot showing the application stack successfully created in this second region.
- **IaC for Business Continuity:** In your own words, explain how having your infrastructure defined as code significantly improves The Riverside Cafe's disaster recovery strategy.

Grading Rubric

Criteria	**Description**	**Points**
IaC Templates	Templates are complete, well-structured, and correctly separated into modular network/application layers. The rationale for modularity is clear.	30
CI/CD Pipeline	The pipeline diagram is accurate. Configuration and successful execution triggered by a code change are proven with screenshots.	30
Cross-Region Deployment	The process for cross-region deployment is described correctly, and screenshot evidence is provided.	20
IaC Justification	The strategic benefit of IaC for disaster recovery is clearly and convincingly explained.	20
Total		**100**